Technical Design Document

1. Project Overview

Radiation Detector Integration with Real-Time Data Recorder

This project is designed to provide an efficient, mobile, and offline-capable radiation monitoring system for technicians operating in nuclear environments. The system consists of a Geiger-Müller tube connected to a Arduino Nano, which transmits real-time radiation readings to an Android tablet application. The app provides an interactive map for logging, storing, and visualizing radiation data.

2. System Architecture

The system follows a modular architecture, consisting of three main components:

A. Hardware Layer

- Geiger-Müller Tube – Detects radiation and generates electrical pulses.
- Arduino Nano– Converts pulses to a digital signal and transmits data to the tablet.
- USB/Wired Connection – Provides secure and reliable communication between the microcontroller and the tablet.

B. Software Layer

- Embedded Firmware (C/C++) – Reads sensor data, filters noise, and formats data for transmission.
- Android Application (Kotlin-based) – Displays real-time radiation readings and logs survey data onto an interactive map.
- Local SQLite Database – Stores survey logs and historical radiation data for offline access.

C. User Interface & Interaction

- Home Screen – Displays live radiation levels.
- Map View – Allows users to drop survey points and visualize radiation intensity.
- Survey Management – Enables users to save, load, and edit surveys.

3. Technology Stack

Component	Technology
Embedded System	C/C++ on Arduino Nano
Data Transmission	USB Serial Communication
Mobile App	Kotlin (Android)
Database	SQLite (Local Storage)
Frontend UI	Jetpack Compose (Kotlin UI Toolkit)

4. Data Flow & Storage

A. Data Flow

1. Radiation Detection – Geiger-Müller tube detects radiation and outputs pulses.
2. Data Processing – Arduino Nano counts pulses and converts them into radiation dose rates.
3. Data Transmission – The microcontroller sends data over a USB connection to the Android tablet.
4. Storage & Visualization – The Android app logs data into SQLite and displays it on a survey map.

B. Data Storage Schema (SQLite)

Table: survey_points

Column	Type	Description
id	INTEGER (Primary Key)	Unique ID for each survey point
location_x	REAL	X-coordinate on the map
location_y	REAL	Y-coordinate on the map
radiation_level	REAL	Measured radiation dose rate
timestamp	DATETIME	When the measurement was taken

5. Functional Requirements

A. Hardware Requirements

- The Geiger-Müller tube must generate pulses that can be counted and processed by the microcontroller.
- The Arduino Nano must maintain a stable wired connection to the tablet.

B. Software Requirements

- The app must display real-time dose rates within ±5% accuracy.
- Users must be able to log survey points by tapping on the map.
- The database must support retrieval and filtering of survey data while offline.

6. User Interface & Interaction

A. Home Screen

- Displays the live radiation level.
- Provides navigation to different features (Maps, Records, Settings).

B. Map View

- Users can tap to drop a survey point and log radiation data.
- Each survey point displays measured dose rate.
- Provides options to edit, delete, and filter surveys.

7. Microcontroller Program Design

1. Receive input from G-M tube (current pulse detection).
2. Maintain a running count per minute and convert it to an exposure rate (Sv/hr).
3. Output exposure rate to the application for real-time display.
4. Transmit signals to the app for recording exposure values.

8. Application Development Plan

A. Setup Home Page with Navigation

- Implement a navigation component to switch between pages.

B. Setup Display and Page Components

- Records Page
- Maps Section:
    - Upload a new map
    - Stored maps for previous records
    - Edit maps for modifications

C. Add Functionalities to Pages

- Ensure survey points can be placed, saved, and edited.
- Ensure maps and records can be retrieved via filtered criteria
- Ensure records can be opened and display the proper survey data

D. Setup Database Schema & Integration

- Connect SQLite database to the application for storage and retrieval.

E. Testing Plan

- Upload a map
- Open a stored map for use
- Record radiation data at survey points
- Save survey data
- Restore a saved map with recorded survey data

9. Future Improvements

- Bluetooth Communication – Replace USB for wireless transmission.
- Cloud Backup – Sync survey data to a secure cloud database.
- AI-Powered Anomaly Detection – Alert users to unusual radiation spikes.
- Local Map Editor – Allow users to create custom maps within the app.
- Data Archiving – Enable users to transfer survey data to an external archive for long-term storage.
- Probe Casing – Design and build a rugged and durable case for the probe